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Roll No

EC-602 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Cellular Mobile Communications
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Describe the parameters responsible for signal loss due to foliage. How does foliage loss vary with foliage density, path lengths and frequency of transmission?
2. Why is FM rather than AM, used in most mobile radio systems today? List as many reasons as you think of and justify your responses. Consider issue such as fidelity, power consumption and noise.
3. With the help of block diagram explain the basic operation of cellular system.
4. Derive the expression for the desired C/I in an Omni-directional antenna system for a cluster size $N=7$.
5. Write short note on the following:
 - i) Umbrella pattern effect.
 - ii) Near and far end interference.
6. Explain the principle working of CDMA system.
OR
Explain what is hand off. Also explain types of hand off techniques.
7. What are the advantages of digital cellular systems over analog? Define PDC, PCN and PCS.
8. Write short notes on any two of the followings
 - i) GSM architecture
 - ii) CDMA
 - iii) TDD systems.
